FINet: A Feature Interaction Network for SAR Ship Object-Level and Pixel-Level Detection

Qi Hu, Shaohai Hu, Shuaiqi Liu, Shuwen Xu, and Yu-Dong Zhang, *Senior Member, IEEE*

***Abstract*— Deep learning-based detection methods have achieved great success in ship target detection in synthetic aperture radar (SAR) images. However, due to the interference of imaging mechanism, speckle noise, and sea and land clutter, ship detection in SAR images still suffers from difficult interpretation. It is found that most ship detection algorithms focus on object-level detection while ignoring pixel-level information. In order to further improve the recognition effectiveness and positioning accuracy of ships in SAR images, we present a novel ship detection method based on a feature interaction network (FINet) in SAR images from the perspective of object-level and pixel-level. FINet consists of an object-level detection network and a pixel-level detection network. The information of the two branches is fused through the feature interaction module (FIM), and then, the object-level information and pixel-level information are enhanced by the feature guidance module (FGM). Finally, FINet utilizes object-level and pixel-level detection heads for prediction and regression to obtain object-level classification accuracy, positioning bounding box coordinates, and pixel-level binary classification results. The experimental results demonstrate that the classification effectiveness and localization accuracy of FINet are better than those of the comparison algorithms, and FINet achieves the best performance.**

***Index Terms*— Convolutional neural network (CNN), multitask learning, ship detection, synthetic aperture radar (SAR) image.**

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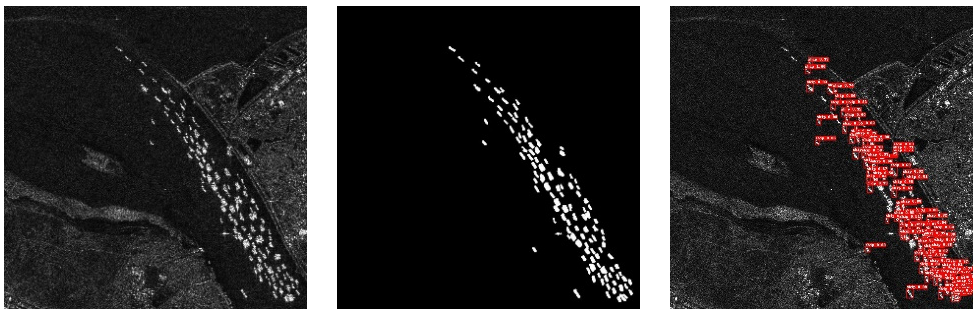
I. Introduction

SYNTHETIC aperture radar (SAR), as an active microwave detection sensor, can provide shareable geospatial intelligence at any time and under any weather conditions, so it plays an important role in military and civil ship detection [1], [2], [3], [4], [5], [6]. In maritime management, SAR can monitor and manage ships entering and leaving ports throughout the day, helping to improve transportation efficiency and reduce maritime traffic accidents; in military affairs, SAR can detect and warn enemy ships, enhance situational awareness, and gain tactical advantages. However, due to the influence of imaging mechanisms, sea and land clutter, and target characteristics, ship detection still has problems in classification effectiveness and positioning accuracy. From the perspective of electromagnetic scattering mechanism, the sea scattering is mainly surface scattering, while the ship target has a different scattering mechanism from the sea surface. The scattering intensity of ship is large, and the scattering intensity of sea surface is small, so it is easy to distinguish ship target from sea clutter. The scattering of islands and land is strong, which has a great impact on ship detection. The traditional ship detection method needs to do sea–land segmentation preprocessing to eliminate the land interference. For deep learning, the features of ships and islands and land are different. Due to the complex structure, the ship target can be considered as a collection of different "angle scatterers." Through the ground truth of specific mark, the network can learn the feature of the ship, so as to eliminate the interference of islands and land. Therefore, ship detection based on deep learning in SAR images is still attracting the attention of researchers.

Traditional SAR image ship detection algorithms are mainly based on pixel-level detection. Since there are only targets such as ships in the image, the usually obtained detection result image is a binary image, as shown in Fig. 1(b), in which white represents the ship target and black represents the background. Pixel-level detection methods mainly include algorithms based on constant false alarm rate (CFAR) [7], [8], [9], [10], [11], [12] and algorithms based on saliency [13], [14], [15], [16], [17]. The detection algorithm based on CFAR is widely used in SAR image ship detection research. CFAR has the advantages of constant false alarm, adaptive threshold, and simple calculation. Novak et al. [7] proposed a two-parameter CFAR algorithm based on Gaussian distribution for ship detection. The parameter estimation of the algorithm is simple, and it has a good detection effect in a uniform background, but the detection performance decreases significantly in a nonuniform

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(a) (b) (c)

Fig. 1. Example images for SAR image ship detection: (a) original image; (b) example of pixel-level detection results; and (c) example of object-level test results.

background; Tao et al. [8] presented a two-parameter CFAR ship detection algorithm based on adaptive truncated clutter statistics, which enhances the detection effect of ships in the nonuniform background; Ai et al. [9] presented a CFAR ship detection method according to truncation statistics, which achieved relatively stable detection performance in the case of high target density. Although the above algorithms have a good detection performance, most of these algorithms require complex parameter settings and are slow, which cannot meet the needs of SAR image information processing.

Saliency-based target detection algorithms usually use local or global contrast to highlight saliency targets to achieve a good target detection performance. Itti et al. [13] designed an Itti visual saliency model according to the visual nervous system of early primates. This method does not need the process of training and learning. It can complete the calculation of saliency map only by the pure mathematical method and achieve a good detection effect; Liu et al. [14] constructed a new SAR image visual attention model according to the Itti model and singular value decomposition (SVD), which improved the detection rate of weak targets in SAR images; Cheng et al. [15] presented a saliency target detection model on the basis of global region contrast. This algorithm generated great saliency feature maps, highlighted the whole target area evenly, and achieved high target detection accuracy; Wang et al. [16] used the idea of pattern recursion to construct a saliency detection model for SAR image target detection in complex background. The detector can suppress the influence of speckle noise and surface texture on target detection, and had strong robustness; Wang et al. [17] presented an unsupervised ship detection method for single-channel SAR images, which achieved high efficiency and high accuracy by using multiscale saliency and complex signal kurtosis. Although the mentioned algorithms have obtained satisfactory target detection effect, these algorithms generally have high complexity, and the detection effect is significantly reduced in complex scenes.

In recent years, scholars have intensively studied deep learning, and object-level target detection methods based on convolutional neural network (CNN) have shown strong performance in ship detection of SAR images [18], [19], [20], [21], [22], [23]. The object-level detection method mainly learns the classification probability and positioning boundary box of the target through neural network, as shown in Fig. 1(c). Yang et al. [18] proposed a new one-stage ship detection method which is called R-RetinaNet. It used feature map scale calibration to align the scale distribution of the output backbone feature map with the target scale distribution, which improved the recognition accuracy of ships; Dai et al. [19] constructed a CNN-based two-stage SAR ship detection model that fused feature maps in a down-top and top-down manner, which makes the location and semantic information of small targets more accurate and improves the performance of ship detection; Cui et al. [20] constructed a new ship detector for large-scale SAR images based on key point estimation. This method can effectively avoid the missed detection of small targets by using CenterNet; Hu et al. [21] built a multiscale balance attention ship detector for SAR images based on anchor-free mechanism, which effectively combined local and nonlocal features to make the model more robust.

The above detection methods have achieved certain results. However, due to the destruction of speckle noise in SAR images and the interference of background clutter, it is difficult for deep networks to learn more general image representations, which is prone to overfitting. In fact, ship detection in a complex background is usually accompanied by multiple tasks, and ship detection is only one of them. Single-task research simply focuses on one task, while multitask research can achieve more representations from different tasks, so that more general features can be learned for better ship detection results. Du et al. [24] presented a new saliency-guided single-shot multibox detector (SSD) in SAR images. The detection algorithm integrated saliency features into the SSD network, which enhances the representation ability of the network in complex scenes and obtains better detection performance on the real dataset. However, the saliency features of the algorithm are directly generated by the improved Itti method, not obtained through network training, so acquired features will be affected by the clutter with high pixel values during network training. Wu et al. [25] presented an instance segmentation-assisted ship detection network (ISASDNet) with two branches. The network improved the result of ship detection through the output interaction of two branches. Zhang et al. [26] built a multitask learning-based target detector (MTL-Det), which modeled the ship detection problem as three collaborative tasks, so as to learn more discriminative target features. Although

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both ISASDNet and MTL-Det have explored ship detection based on multitask learning, each subtask in ISASDNet and MTL-Det only shared the initial features of the network, and each learned deep features independently, which is not conducive to learning the similarities and differences between tasks.

By exploring the above problems, in this study, we consider the interaction between different subtasks at different scales and present a novel ship detection algorithm based on a feature interaction network (FINet) in SAR images from the perspective of object-level and pixel-level. Through the constructed feature interaction module (FIM), FINet performs a multiscale interactive fusion of the information of object-level and pixel-level branches to get deep multiscale feature about SAR image. Then, FINet uses a feature guidance module (FGM) to enhance object-level information and pixel-level information, respectively, to obtain a more robust detection network. More specifically, the main contributions of our work are summarized as follows.

1) In this study, a novel ship detection method for SAR images based on a FINet is present. This algorithm utilizes the deep network to simultaneously study the multitask and multiscale features of the target, and obtains a better ship detection effect. The experimental results on high-resolution SAR images dataset (HRSID) and SAR ship detection dataset (SSDD) demonstrate that the present algorithm has a high ship detection accuracy and a good generalization ability.
2) We construct a FIM, which interacts with object-level features and pixel-level features on different scales, so that the proposed network can obtain deeper and more general representations of the images. We also use the FGM to fuse and enhance the features of the two subtasks to improve the robustness of the detection algorithm.
3) In this article, two detection heads of object-level and pixel-level branches are designed for prediction and regression, which can simultaneously obtain object-level classification accuracy, localization bounding box coordinates, and pixel-level binary classification results that are more in line with the task of SAR image ship detection.

The rest parts of this article are organized as follows. Section II introduces FINet in detail. The experimental results and analysis are presented in Section III. Section IV gives the conclusion.

II. Methodology

Most SAR image ship target detection focuses on the object-level feature of the ship, and the main task is to classify the ship and regress the specific location of the target, while ignoring the pixel-level information. Pixel-level detection can not only provide clear detection results that are more consistent with visual perception when ships are dense, but also provide more detailed positioning information for object-level detection. Therefore, in this study, we present a ship detection method for SAR image based on a FINet. The structure of FINet is shown in Fig. 2. FINet is composed of two subtask branches: object-level branch and pixel-level branch. It can enhance the interaction and integration between object-level feature and pixel-level feature in different branches and scales, so as to achieve the accurate detection of ship targets. FINet is also equipped with object-level and pixel-level detection heads, which can obtain object-level classification accuracy, positioning bounding box coordinates, and pixel-level binary classification results. The main components of FINet are introduced in detail in the following.

A. Feature Interaction Module

Multitask learning has been extensively studied and discussed in target detection. Scholars use multitask learning to obtain more feature representations of input data from different tasks, so as to make the network more universal. However, most methods only add another task feature to one task or each subtask only shares the initial feature of the network, while each learns the deep features of the data independently, which is not conducive to learning the similarities and differences between tasks. In order to obtain more general features, the FIM is constructed in this article. The two subtasks of object-level detection and pixel-level detection are used for interactive learning, so as to obtain more discriminative target features. Before the interaction of depth features, we use ResNet [27] to build a backbone network to obtain the initial multiscale features B_1, B_2, B_3, B_4, and B_5 of the input data, where B_2, B_3, B_4, and B_5 will be input to the two subtask branches of object-level and pixel-level for interactive learning. A detailed introduction is given in the following.

First, we utilize the initial feature B_5 to obtain object-level feature O_5 and pixel-level feature P_5. O_5 and P_5 are calculated as follows:

$$O_5 = \text{Conv}_{3\times3}(\text{Conv}_{1\times1}(B_5)) \tag{1}$$

$$P_5 = \text{Conv}_{3\times3}(\text{Conv}_{1\times1}(B_5)) \tag{2}$$

where $\text{Conv}_{1\times1}$ means 1×1 convolution, which is used for channel compression. $\text{Conv}_{3\times3}$ represents 3×3 convolution, which is used to extract image features.

Then, we input the initial feature B_l, object-level feature O_l, and pixel-level feature P_l ($l = 3,4,5$) into FIM for interactive learning. The FIM structure is shown in Fig. 3.

In Fig. 3, the object-level feature O_l and pixel-level feature P_l are stacked as S_l in the channel dimension, that is,

$$S_l = O_l \copyright P_l \tag{3}$$

where ©indicates the concat function. The number of channels for both O_l and P_l is C, and the number of channels of S_l is $2C$. In order to get the shared representation of the two branch features, BasicBlocks1 is first used to carry out a learnable nonlinear mapping with the number of output channels of $2C$. Second, the output is divided into two feature groups along the channel dimension, and the number of channels in each group is C. Then, in this article, we apply the Sigmoid function along the task dimension to generate a task attention mask to extract the structural features of S_l. The mask is further processed by BasicBlocks2 to decrease the number of channels from $2C$ to C, thus obtaining the shared representation F_l of

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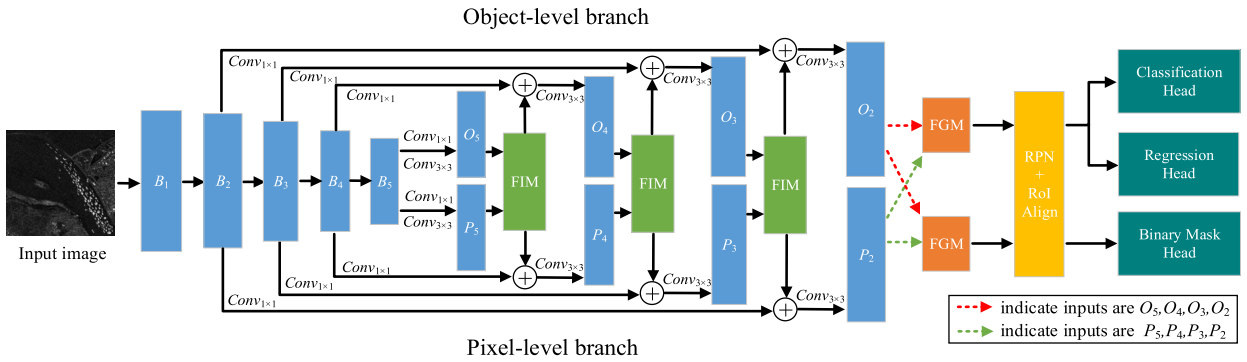


Fig. 2. Framework of FINet. FINet consists of two subtask branches: object-level branch and pixel-level branch. Object-level branch is to obtain object-level classification accuracy and positioning boundary box coordinates. Pixel-level branch is to obtain pixel-level binary classification results.

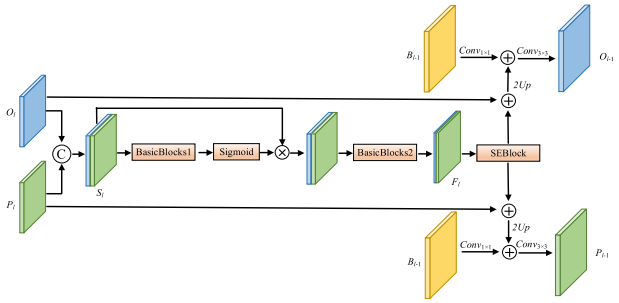


Fig. 3. Structure of FIM.

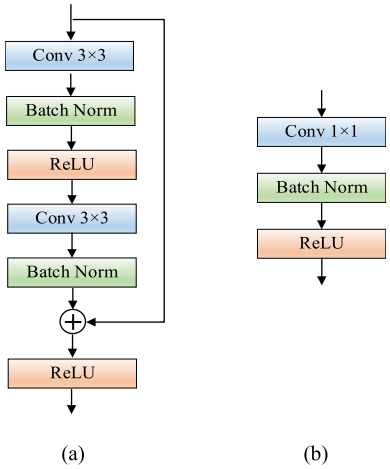


Fig. 4. Structure of BasicBlocks: (a) BasicBlock and (b) ConvBlock.

object-level feature and pixel-level feature. The whole process is shown as follows:

$$F_l = \text{BasicBlocks2}(\text{Sigmoid}(\text{BasicBlocks1}(S_l)) \otimes S_l) \quad (4)$$

where $\otimes$ represents the dot multiplication operation. BasicBlocks1 consists of three BasicBlock. The BasicBlock structure is shown in Fig. 4(a), which includes a residual branch and a shortcut branch. BasicBlock is mainly composed of two 3×3 convolution layers, batch normalization layer, and rectified linear unit (ReLU) activation operation. Compared with the traditional convolution structure, there is one more shortcut branch for transmitting low-level information, so that the deep network can effectively avoid the phenomenon of gradient disappearance. BasicBlocks2 consists of two BasicBlock and one ConvBlock. The ConvBlock structure is shown in Fig. 4(b), which includes a 1×1 convolution operation, batch normalization layer, and ReLU activation layer. The 1×1 convolution operation can reduce the number of channels of the current feature map from $2C$ to C, so as to obtain shared features F_l that match the initial features B_l, object-level features O_l, and pixel-level features P_l.

When tasks are irrelevant, using shared representation may reduce the performance of the network. To solve this problem, we select information from the shared representation through the sequence and excitation block (SEBlock) [28]. That is, each subtask can select relevant features from the shared representation and add them to the original features O_l and P_l as residuals. Finally, connecting with the previous higher resolution backbone features B_{l-1} and the selected task features, we can obtain the object-level features O_{l-1} and pixel-level features P_{l-1} containing the interaction information of the two branches. The whole process is calculated as follows:

$$O_{l-1} = \text{Conv}_{3\times3}(2\text{Up}(\text{SEBlock}(F_l) \oplus O_l) \oplus \text{Conv}_{1\times1}(B_{l-1})) \quad (5)$$

$$P_{l-1} = \text{Conv}_{3\times3}(2\text{Up}(\text{SEBlock}(F_l) \oplus P_l) \oplus \text{Conv}_{1\times1}(B_{l-1})) \quad (6)$$

where $\oplus$ refers to the point-to-point addition operation. 2Up refers to the twofold upsampling operation, which is used to upsample the current feature to the same scale as the upper level backbone feature. When l takes 3, 4, and 5, we can obtain object-level features O_4, O_3, O_2 and pixel-level features P_4, P_3, P_2 after interacting at different scales for different tasks.

B. Feature Guidance Module

Through the FIM, we carry out multiscale interaction between the information of object-level and pixel-level detection tasks, which not only adds deep supervision to the network, but also realizes the extraction of task features at each level. In order to further refine the features, we construct an uncomplicated FGM to obtain the attention features of the two branches. Fig. 5 shows the structure diagram of FGM.

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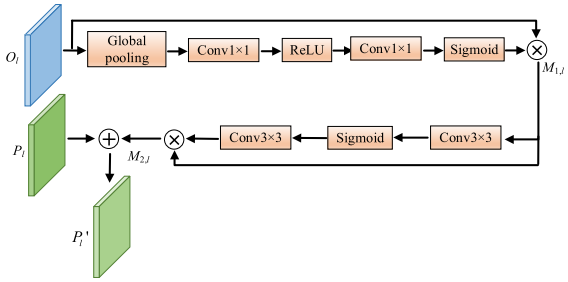


Fig. 5. Structure diagram of FGM.

First, the object-level feature O_l passes through a global pooling unit to pool the feature map size from $H \times W \times C$ to $1 \times 1 \times C$. Second, the complex correlation between channels is fit by two 1×1 convolutional layers to downscale and then upscale the feature map, adding more nonlinear processing. Then, the feature map is passed through a Sigmoid layer to obtain a $1 \times 1 \times C$ channel attention mask. Finally, we multiply the channel attention mask with the input O_l to obtain the channel attention feature $M_{1,l}$, that is,

$$M_{1,l} = O_l \otimes \text{Sigmoid}(\text{Conv}_{1\times1}(\text{ReLU}(\text{Conv}_{1\times1}(\text{GP}(O_l))))) \tag{7}$$

where GP represents the global pooling operation, $\otimes$ represents the dot product operation, and ReLU represents the activation function.

Then, in order to obtain the spatial attention feature of O_l, 3×3 convolution operation and Sigmoid operation are performed on $M_{1,l}$ to obtain attention feature $M_{2,l}$, that is,

$$M_{2,l} = M_{1,l} \otimes \text{Conv}_{3\times3}\big(\text{Sigmoid}\big(\text{Conv}_{3\times3}\big(M_{1,l}\big)\big)\big). \tag{8}$$

Feature $M_{2,l}$ contains the channel attention and spatial attention features of input object-level features O_l. Finally, $M_{2,l}$ is added to the feature P_l to obtain pixel-level features P_l' after feature enhancement, that is,

$$P_l' = P_l \oplus M_{2,l}. \tag{9}$$

Similarly, the enhanced object-level features O_l' can be obtained as follows:

$$S_{1,l} = P_l \otimes \text{Sigmoid}(\text{Conv}_{1\times1}(\text{ReLU}(\text{Conv}_{1\times1}(\text{GP}(P_l))))) \tag{10}$$

$$S_{2,l} = S_{1,l} \otimes \text{Conv}_{3\times3}\big(\text{Sigmoid}\big(\text{Conv}_{3\times3}\big(S_{1,l}\big)\big)\big) \tag{11}$$

$$O_l' = O_l \oplus S_{2,l} \tag{12}$$

where $S_{1,l}$ is the channel attention of the pixel-level feature P_l. $S_{2,l}$ contains the channel attention and spatial attention features of P_l, and O_l' is the object-level feature containing the pixel-level attention.

Through the FGM module, the proposed network obtains the attention features of object-level task and guides them to the pixel-level task. At the same time, the network takes the attention features for the pixel-level task and guides them to the object-level task. Thus, enhanced pixel-level features and object-level features are obtained. Section II-C will describe how to use these features to make the final detection head prediction.

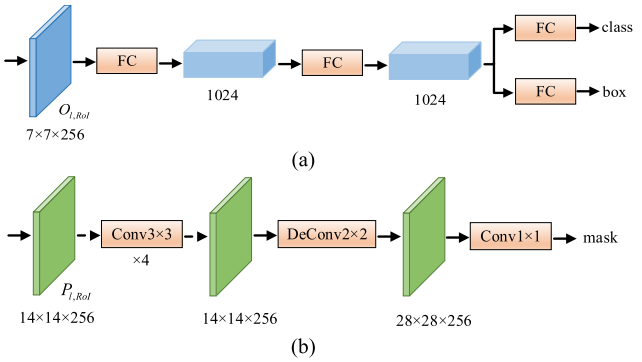


Fig. 6. Structure of detection heads: (a) structure diagram of the classification head and regression head and (b) structure of the mask head.

C. Detection Heads

FINet includes three detection heads: the classification head for calculating the classification accuracy of ships, the regression head for positioning the coordinates of the bounding box, and the binary mask head. Fig. 6 gives the specific structure diagram. FINet adopts a two-stage detection method for ship detection. The structure of detection heads in this article is drawn from the Mask region-CNN (R-CNN) [29]. O_l' is aligned to feature $O_{l,\text{RoI}}$ after region proposal network (RPN) and region of interest (RoI) align operations, and $O_{l,\text{RoI}}$ is used as the input of the classification head and regression head. Then, two full connection layers are used to convert the 2-D feature map into a 1-D vector, that is, the feature space is mapped to the sample label space. Finally, the prediction category and the prediction boundary box coordinates are obtained by using the full connection layers. P_l' is aligned to feature $P_{l,\text{RoI}}$ after RoI align operation, and $P_{l,\text{RoI}}$ is used as the input of the mask head. Then, the features of the detection target are learned through four 3×3 convolution layers, and the feature map size is changed from $14 \times 14 \times 256$ to $28 \times 28 \times 256$ by deconvolution operation to obtain a more detailed mask. Finally, 1×1 convolution layer is used to get the prediction mask of the ship. Through the classification head, regression head, and mask head, the object-level and pixel-level detection results are predicted, and the object-level classification accuracy, positioning bounding box coordinates, and pixel-level binary classification results are obtained.

D. Loss Function

In order to make the constructed network converge faster, the whole loss function is composed of classification loss, regression loss, and mask loss, and we believe that these three parts have the same contribution to ship detection. Therefore, the training loss function in this article is as follows:

$$L = L_{\text{cls}} + L_{\text{reg}} + L_{\text{mask}} \tag{13}$$

where L_{cls} represents the classification loss, which is defined by logarithmic loss, that is,

$$L_{\text{cls}} = \frac{1}{N_{\text{cls}}} \sum_i -\log\left[p_i^* p_i + \left(1 - p_i^*\right)(1 - p_i)\right] \tag{14}$$

where N_{cls} represents the total number of positive and negative samples, i represents the index of each anchor, p_i represents

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the probability that the network predicts the sample to be the target, and p_i^* is the truth label, which takes a value of 1 or 0.

L_{reg} represents the regression loss, which is defined by L1 loss, that is,

$$L_{\text{reg}} = \frac{1}{N_{\text{reg}}} \sum_i p_i^* \text{smooth}_{L1}\left(t_i - t_i^*\right) \tag{15}$$

where N_{reg} represents the total number of positive and negative samples, p_i^* ensures that regression loss is activated when the sample is positive (i.e., $p_i^* = 1$), and

$$\text{smooth}_{L1}\left(t_i - t_i^*\right) = \begin{cases} 0.5\left(t_i - t_i^*\right)^2, & \text{if } \left|t_i - t_i^*\right| < 1 \\ \left|t_i - t_i^*\right| - 0.5, & \text{otherwise} \end{cases}$$

t_i expresses the predicted offset of the anchor box during the training phase. t_i^* expresses the real offset of the anchor box relative to the ground truth during the training phase.

L_{mask} [29] represents the mask loss, which is obtained as follows:

$$L_{\text{mask}} = -\frac{1}{N_{\text{mask}}} \sum_j y_j^* \log y_j + \left(1 - y_j^*\right) \log\left(1 - y_j\right) \tag{16}$$

where N_{mask} represents the total number of pixels in the output feature map of the mask branch, y_j represents the segmentation result of the jth pixel, and y_j^* represents the real segmentation result of the jth pixel.

III. Experimental Results and Analysis

A. Datasets and Training Settings

We adopt HRSID [30] and SSDD [31], [32] datasets for training and testing to verify the performance of the proposed method. These two datasets not only give the ship's coordinates and classification results, but also give the ship's pixel-level results, that is, the binary information of the image. HRSID contains 5604 SAR images and 16 951 ship targets. The image size is 800 × 800 and the resolution is 0.5–3 m. The annotation information is the horizontal box and pixel-level ground truth. Fig. 7(a) shows some images of HRSID. We take 65% of the images as the training set and 35% as the test set. SSDD dataset was released earlier, and the annotation information is horizontal box, rotating box, and pixel-level polygon segmentation, which has been confirmed by professional interpreters. This dataset contains 1160 images with different sizes and resolutions of 1–15 m. Fig. 7(b) shows some images of SSDD. SSDD dataset further divides the images into offshore and inshore types. In this article, the detection algorithms are trained and tested under these two scene types to verify the detection effect of these algorithms under simple samples (offshore scene) and difficult samples (inshore scene).The proposed FINet uses the weight of ResNet-101 model pretrained on ImageNet [33] to initialize the backbone network. In our study, the weight decay is 0.0001, and the momentum is 0.9. The initial learning rate is 0.001, and the batch size is set to 2. For the HRSID dataset, FINet is trained for 24 epochs, and the learning rate drops by a factor of 0.1 at 16th and 22nd epochs. For the SSDD dataset, FINet is trained for 12 epochs, and the learning rate drops by a factor of 0.1 at 8th and 11th epochs. The experimental platform is a personal computer with a Ubuntu 16.04 Linux operating system and a GeForce RTX 2080Ti GPU card. The deep learning framework used in this algorithm is PyTorch.

TABLE I
Results of Ablation Experiment

	Precision	*Recall*	*F1*	AP_{50}	AP_{75}	AP_s
Case1	0.586	0.629	0.607	0.834	0.652	0.592
Case2	0.652	0.690	0.670	0.899	0.745	0.660
Case3	0.525	0.605	0.562	0.788	0.612	0.558
Case4	0.660	0.699	0.679	0.884	0.769	0.670
Case5	0.531	0.610	0.568	0.798	0.618	0.565
Case6	**0.686**	**0.720**	**0.703**	**0.905**	**0.796**	**0.691**

B. Evaluation Index

We use precision (Precision), recall (Recall), F1 score (F1), and average precision (AP) to measure the ship detection performance of the detection algorithms on HRSID and SSDD datasets. The objective indices can be calculated by

$$\text{Precison} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \tag{17}$$

$$\text{Recall} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{18}$$

$$\text{F1} = \frac{2 \times \text{Precison} \times \text{Recall}}{\text{Precison} + \text{Recall}} \tag{19}$$

$$\text{AP} = \frac{1}{101} \sum_{r \in S} \text{Precison}|_{\text{Recall}=r} \tag{20}$$

where N_{TP}, N_{FP}, and N_{FN} indicate the number of true positives, false positives, and false negatives, respectively. Precision indicates the proportion of correctly detected ships in all predictions. Recall indicates the proportion of positive ships over all ground truths. The F1 value is a comprehensive indicator, and the higher the F1, the better the overall detection performance. AP is a classical measure metric for target detection, which is described as the AP under a group of equidistant recalls $S = [0,\ 0.01, \ldots, 1]$. The average accuracy can be divided into AP, AP_{50}, AP_{75}, AP_s, AP_m, and AP_l according to different intersection over union (IoU) and different measurement standards [34]. Since most of the ships in the SAR image belong to small targets, we select AP_{50}, AP_{75}, and AP_s from the six AP indicators to test the algorithm.

C. Ablation Experiment

We use the HRSID dataset to conduct ablation experiments to explore the role of each component of the FINet, and set up six cases: Case1 expresses Faster R-CNN whose backbone is ResNet and has no neck; Case2 represents Faster R-CNN with residual network (ResNet) and feature pyramid network (FPN); Case3 indicates that the network includes two branches of object-level and pixel-level detection but does not include FIM and FGM; Case4 represents the network structure of adding FIM in Case3; Case5 represents the structure of adding FGM in Case3; and Case6 represents the complete FINet constructed in this article. Table I shows the ablation study results. We can observe that Case1 has the worst effect. The

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accuracy of Case2 based on ResNet and FPN at the same time has been greatly improved than that of Case1, but it is still worse than that of Case4 by using ResNet and FIM. This shows that the FIM constructed in this article can better perform multiscale and multitask feature interaction, so the detection accuracy is better than that of using only FPN for multiscale information extraction. The result of Case4 is better than that of Case3, and the result of Case6 is better than that of Case5, which fully proves the effectiveness of the FIM. FIM can fully interact the information of different branches and can substantially improve the representation ability of the model. The result of Case5 is better than that of Case3, and the result of Case6 is better than that of Case4, which proves that FGM can further enhance the information of different branches, thus further improving the detection results.

D. Experimental Results on HRSID Dataset

In this section, we compare FINet with traditional and classic deep learning ship detection methods on the HRSID dataset to demonstrate the effectiveness of the proposed method. The comparison algorithms include the following.

1) The target detection method based on Darknet-53 proposed in [35], which is referred to as YOLOv3.
2) The one-stage detection algorithm proposed in [36] by using focal loss optimization training, which is referred to as RetinaNet.
3) The two-stage target detection algorithm proposed in [37], which is called Faster R-CNN for short.
4) The two-stage target detection algorithm based on different IoU thresholds proposed in [38], which is called Cascade R-CNN for short.
5) The ship detection algorithm for SAR image based on the FPN proposed in [39], which is referred to as HR-SDNet.
6) The one-stage target detection and segmentation method proposed in [40], which is called Yolact for short.
7) The two-stage target detection and segmentation method proposed in [29], which is called Mask R-CNN for short.
8) The target detection method based on visual attention presented in [41], referred to as VAB.
9) The target detection method based on super-pixel CFAR presented in [42], referred to as SP-CFAR.

Among these algorithms, YOLOv3, RetinaNet, Faster R-CNN, Cascade R-CNN, and HR-SDNet are object-level target detection algorithms, while VAB and SP-CFAR are pixel-level detection algorithms, and Yolact and Mask R-CNN are both object-level and pixel-level detection algorithms. For the sake of matching the results of the traditional ship pixel-level detection algorithms, the effect map of the deep learning method is processed into a binary image in this article.

1) Object-Level Detection Results: The object-level detection algorithms are used for testing on the HRSID dataset, and Table II lists the detection results. We can observe that the overall performance of the one-stage detection algorithms YOLOv3, RetinaNet, and Yolact is poor, not as good as the two-stage algorithms. The detection indicators of the two-stage algorithms are generally high. Mask R-CNN has the highest

TABLE II
EVALUATION INDEXES OF EACH OBJECT-LEVEL DETECTION METHOD ON HRSID

	Precision	*Recall*	*F1*	AP_{50}	AP_{75}	AP_s
YOLOv3	0.602	0.652	0.626	0.904	0.684	0.612
RetinaNet	0.599	0.650	0.623	0.855	0.664	0.606
Faster R-CNN	0.652	0.690	0.670	0.899	0.745	0.660
Cascade R-CNN	0.662	0.705	0.682	0.885	0.753	0.675
HR-SDNet	0.651	0.693	0.671	0.896	0.749	0.664
Yolact	0.564	0.616	0.589	0.846	0.630	0.577
Mask R-CNN	0.677	0.714	0.695	**0.912**	0.781	0.679
FINet	**0.686**	**0.720**	**0.703**	0.905	**0.796**	**0.691**

TABLE III
EVALUATION INDEXES OF PIXEL-LEVEL DETECTION METHODS ON HRSID

	Precision	*Recall*	*F1*	AP_{50}	AP_{75}	AP_s
Yolact	0.334	0.382	0.356	0.688	0.283	0.312
Mask R-CNN	0.563	0.607	0.584	**0.889**	0.670	0.553
FINet	**0.580**	**0.619**	**0.599**	0.886	**0.698**	**0.570**

performance among all the comparison algorithms, but it is still lower than FINet. FINet has the best precision and recall. In addition, the F1 index is the highest, demonstrating that the comprehensive performance of the proposed algorithm is better. The AP_{50} of FINet is slightly lower than the corresponding value of Mask R-CNN, but the more stringent AP_{75} index is 1.5% higher than that of Mask R-CNN. This shows that FINet can obtain better quality bounding boxes and achieve more accurate target positioning. The proposed algorithm also obtains the optimal value on the AP_s index, which fully proves that FINet has higher detection accuracy for small targets. In order to better illustrate the advantages of FINet, the corresponding precision-recall (PR) curve is drawn in Fig. 8. It can be found that FINet (the red curve) has the largest area under the curve, which proves that FINet has better ship detection performance.

In addition, we randomly select an SAR image from the HRSID dataset for visualization. The ships in this SAR image are in an inshore environment and are susceptible to interference from land scattering. In addition, many ships in many places are closely connected, so detection is difficult. Fig. 9 gives the detection results. Fig. 9(a) presents the ground truth (ships are framed with red boxes), and the remaining images show the detection results of different algorithms. The red circle indicates false detection, and yellow circle represents missed detection. YOLOv3, RetinaNet, and Yolact have multiple missed detections; Faster R-CNN and Cascade R-CNN have multiple false detections and missed detections; HR-SDNet and Mask R-CNN both have two false detections and one missed detection; and FINet has only one false detection and one missed detection. There are two ships in Region I [marked by the green box in Fig. 9(a)] head-to-end. Only FINet detects these two ships more accurately, and the confidence levels are 0.86 and 0.85. In summary, the proposed algorithm has better ship detection performance and can more accurately detect ships in lose distances and ships in inshore environments.

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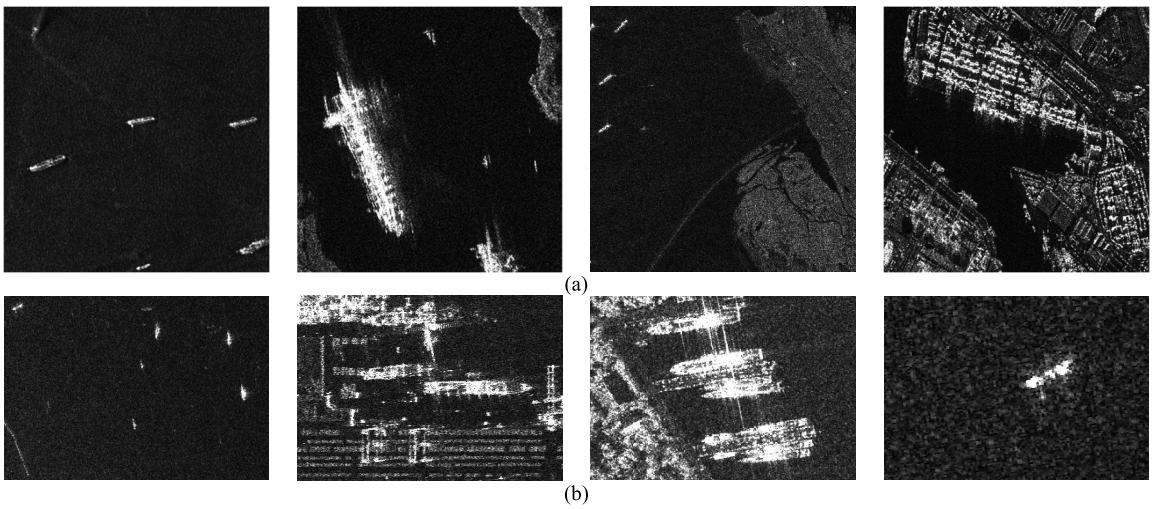


Fig. 7. Sample images in HRSID and SSDD: (a) images in HRSID and (b) images in SSDD.

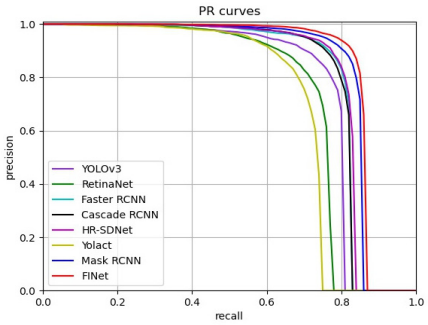


Fig. 8. PR curve of each object-level detection method on HRSID.

TABLE IV
EVALUATION INDEXES OF OBJECT-LEVEL DETECTION METHODS FOR OFFSHORE TARGETS ON SSDD

	Precision	*Recall*	*F1*	AP_{50}	AP_{75}	AP_s
YOLOv3	0.563	0.637	0.598	0.973	0.595	0.566
RetinaNet	0.709	0.761	0.734	0.976	0.878	0.708
Faster R-CNN	0.725	0.768	0.746	0.990	**0.926**	0.729
Cascade R-CNN	0.738	0.775	0.756	0.988	0.879	0.735
HR-SDNet	0.743	0.786	0.764	0.989	0.908	0.741
Yolact	0.357	0.476	0.408	0.871	0.179	0.416
Mask R-CNN	0.736	0.776	0.755	0.989	0.903	0.748
FINet	**0.757**	**0.796**	**0.776**	**0.998**	0.923	**0.765**

2) Pixel-Level Detection Results: In this section, we use HRSID dataset to test the pixel-level detection algorithms, and Table III lists the detection results. We can see that the one-stage detection algorithm Yolact has the worst results and cannot extract pixel-level information well. The indicators of the two-stage detection method Mask R-CNN are better, but there is still a gap compared with FINet. Among all the detection algorithms, FINet has the best precision, recall, and F1 score, which fully shows that our presented algorithm has optimal detection accuracy and good comprehensive performance. FINet's AP_{50} is only 0.3% lower than Mask R-CNN, while its AP_{75} and AP_s are 2.8% and 1.7% higher than Mask R-CNN, respectively. We can conclude that FINet can extract the target features more accurately and achieve satisfactory results in pixel-level detection.

Similarly, we randomly select an SAR image from the HRSID dataset to show the effect of each pixel-level detection algorithm. There is a great deal of dense small ships in the selected SAR image, so detection is difficult. Especially when using object-level annotation, the objects with a short distance are easily covered up [as shown in Fig. 1(c)], which makes the detection algorithm unable to accurately interpret the image. Fig. 10 gives the detection results. Fig. 10(a) shows the original SAR image, Fig. 10(b) shows the ground truth, and Fig. 10(c)–(f) shows the experimental result of different methods. Red circles indicate false detections, and yellow circles indicate missed detections. VAB uses the global saliency of the image for detection, so it is greatly affected by land clutter. VAB misses many targets and has the poor detection effect. SP-CFAR is a classic CFAR algorithm, but its detection results still inevitably leave land clutter and some targets are missing. Mask R-CNN and FINet have better detection results, but FINet has fewer false detections and missed detections than Mask R-CNN. It shows that FINet effectively avoids clutter interference and preserves the ship target relatively completely.

E. Experimental Results on SSDD Dataset

The presented algorithm is trained on SSDD dataset and tested in both offshore and inshore conditions to prove its generalization performance. The comparison algorithms used are the same as in Section III-D.

1) Object-Level Detection Results: The object-level detection indicators of offshore targets on the SSDD dataset are shown in Table IV. Similar to the test effect on HRSID, Table IV lists that the detection results of the one-stage detection algorithms (YOLOv3, RetinaNet, and Yolact) are poor, and the accuracy rates are all low. The indexes of the two-stage detection algorithms are better, and the accuracy rate is higher than 70%. HR-SDNet has the best overall performance among

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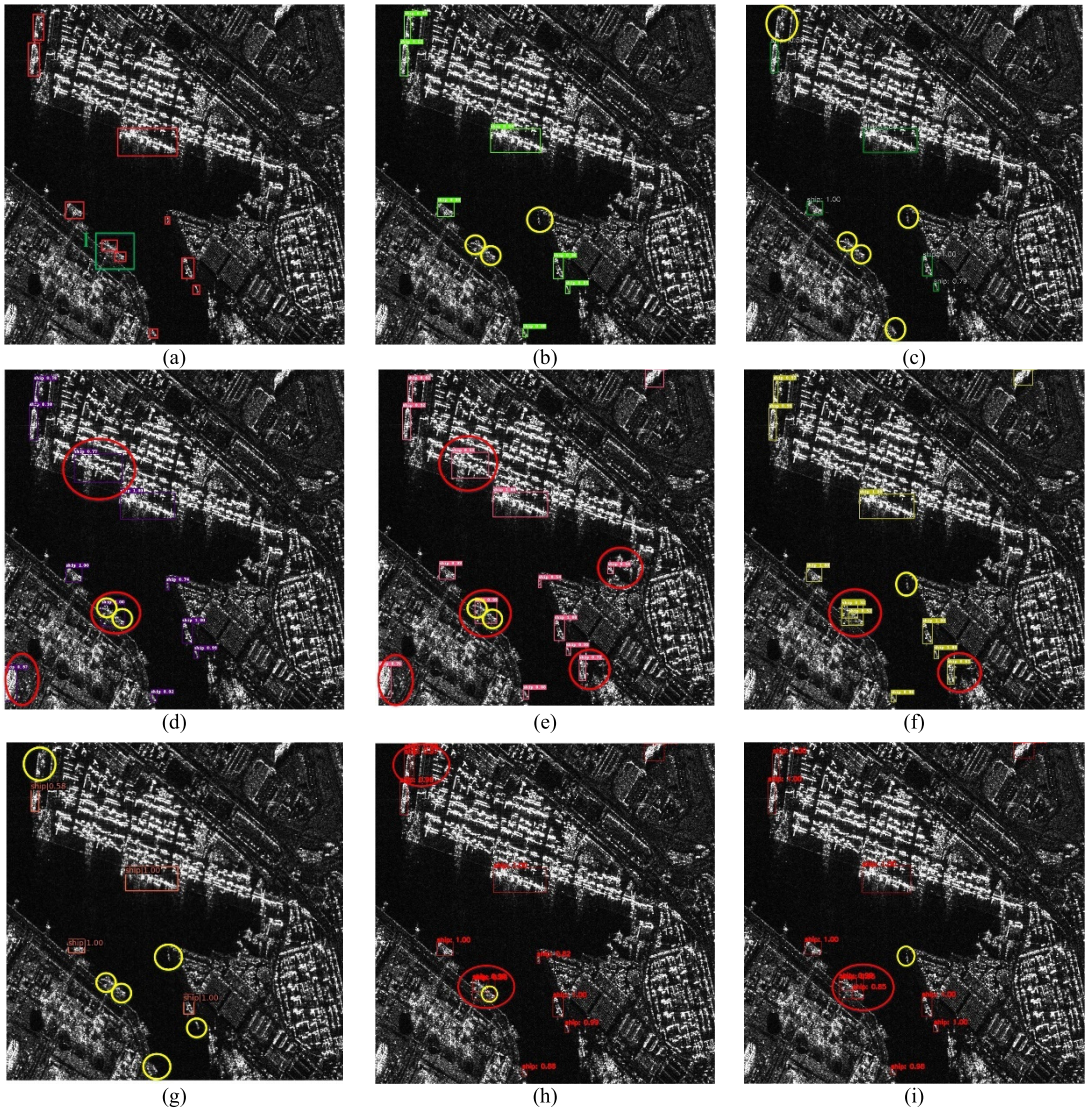


Fig. 9. Object-level detection results of different algorithms on HRSID: (a) ground truth; (b) YOLOv3; (c) RetinaNet; (d) Faster R-CNN; (e) Cascade R-CNN; (f) HR-SDNet; (g) Yolact; (h) Mask R-CNN; and (i) FINet.

all the comparison algorithms. However, FINet achieves the highest precision and recall among the eight algorithms, and its F1 index is also the highest. The AP_{75} of FINet is only a little lower than that of Faster R-CNN, but the values of AP_{50} and AP_s are significantly higher than those other algorithms, which fully proves the excellent comprehensive performance of FINet. Fig. 11 illustrates the PR curves of all algorithms. In Fig. 11, the red curve represents FINet. It can be observed that FINet has the largest area under the curve, which shows that the proposed model has better ship detection performance.

In addition, we randomly select an SAR image with offshore targets from the SSDD dataset for visualization. The SAR image is shown in Fig. 12. Fig. 12 shows that there is a large amount of speckle noise, which seriously interferes with ship detection. Fig. 12(a) shows the ground truth (the two ships are marked A and B with red boxes, respectively), and the rest of the subimages are the detection results of each algorithm. We can see that Yolact misses ship A and falsely detects large speckle noise as a ship target. HR-SDNet falsely detects two ships as one. The rest algorithms detect the target correctly, but only Faster R-CNN and FINet had a target confidence of 1.00. This shows that the proposed algorithm has a stronger antinoise ability and can robustly detect and locate targets under high noise conditions.

The object-level detection indicators of inshore targets on the SSDD dataset are shown in Table V. The inshore ship targets are disturbed by land clutter, and the detection is more difficult than that of offshore ships. Therefore, compared with Table IV, the detection accuracy of each detection algorithm is decreased and the performance is unbalanced. Table V illustrates that the precision of the one-stage detection methods is all lower than 50%. The indicators of the two-stage detection algorithms are generally better, and HR-SDNet performs best in all comparison algorithms. FINet is only slightly lower than Mask R-CNN on the AP_{75} and slightly lower than HR-SDNet on the AP_s. But it has the highest precision,

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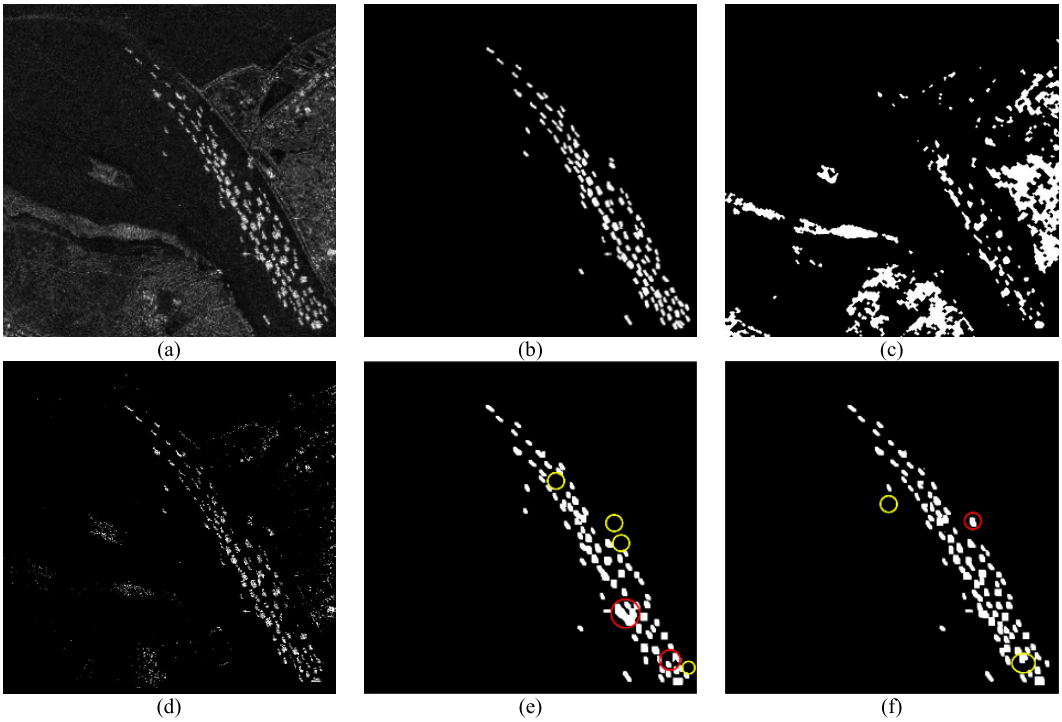


Fig. 10. Pixel-level detection results of different algorithms on HRSID: (a) original image; (b) ground truth; (c) VAB; (d) SP-CFAR; (e) Mask R-CNN; and (f) FINet.

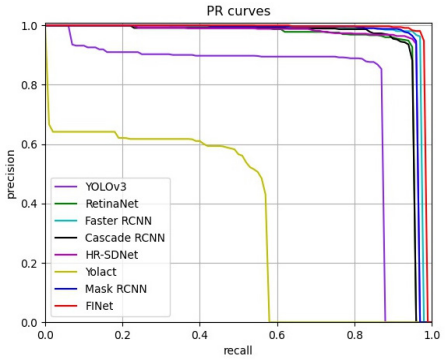


Fig. 11. PR curve of object-level detection methods for offshore targets on SSDD.

recall, comprehensive performance index F1, and AP_{50}, which demonstrates that FINet has better ship detection performance.

Fig. 13 illustrates the PR curves of each algorithm. Compared with the PR curve of the offshore target in Fig. 11, the area under the curve of each algorithm is reduced, which shows that the detection of the inshore target is more difficult and the detection effect is worse. In Fig. 13, the red curve represents FINet. It can be observed that FINet has the largest area under the curve. Combining Table V and Fig. 13, we can conclude that FINet is more robust and has better comprehensive performance.

We randomly select an SAR image containing inshore targets from the SSDD dataset for visualization to demonstrate the performance of our algorithm. The SAR image is shown in Fig. 14. There are not only inshore targets, but also two ships docked together, which are difficult to distinguish. Fig. 14(a) shows ground truth (three ships are marked with green boxes as A, B, and C, respectively), and Fig. 14(b)–(i) shows the detection results of each algorithm. Fig. 14 shows that RetinaNet, Mask R-CNN, Cascade R-CNN, and Yolact fail to correctly detect ships A and B, and Mask R-CNN misses ship C. YOLOv3, Faster R-CNN, and HR-SDNet fail to detect ship A. Only FINet can accurately detect ship A, which shows that FINet can effectively distinguish closely arranged targets, and its performance is better than other comparison algorithms.

TABLE V
EVALUATION INDEXES OF OBJECT-LEVEL DETECTION METHODS FOR INSHORE TARGETS ON SSDD

	Precision	*Recall*	*F*1	AP_{50}	AP_{75}	AP_s
YOLOv3	0.463	0.540	0.499	0.893	0.410	0.516
RetinaNet	0.469	0.567	0.513	0.747	0.508	0.523
Faster R-CNN	0.548	0.602	0.574	0.839	0.650	0.605
Cascade R-CNN	0.546	0.614	0.578	0.829	0.613	0.591
HR-SDNet	0.570	**0.635**	0.601	0.889	0.599	**0.618**
Yolact	0.191	0.345	0.246	0.434	0.115	0.307
Mask R-CNN	0.557	0.626	0.589	0.876	**0.666**	0.608
FINet	**0.573**	**0.635**	**0.602**	**0.905**	0.651	0.598

TABLE VI
EVALUATION INDEXES OF PIXEL-LEVEL DETECTION METHODS FOR OFFSHORE TARGETS ON SSDD

	Precision	*Recall*	*F1*	AP_{50}	AP_{75}	AP_s
Yolact	0.501	0.607	0.549	0.904	0.510	0.542
Mask R-CNN	0.696	0.734	0.714	0.989	0.920	0.690
FINet	**0.711**	**0.751**	**0.730**	**0.998**	**0.926**	**0.706**

2) Pixel-Level Detection Results: Table VI lists the evaluation indicators of pixel-level detection of offshore targets on

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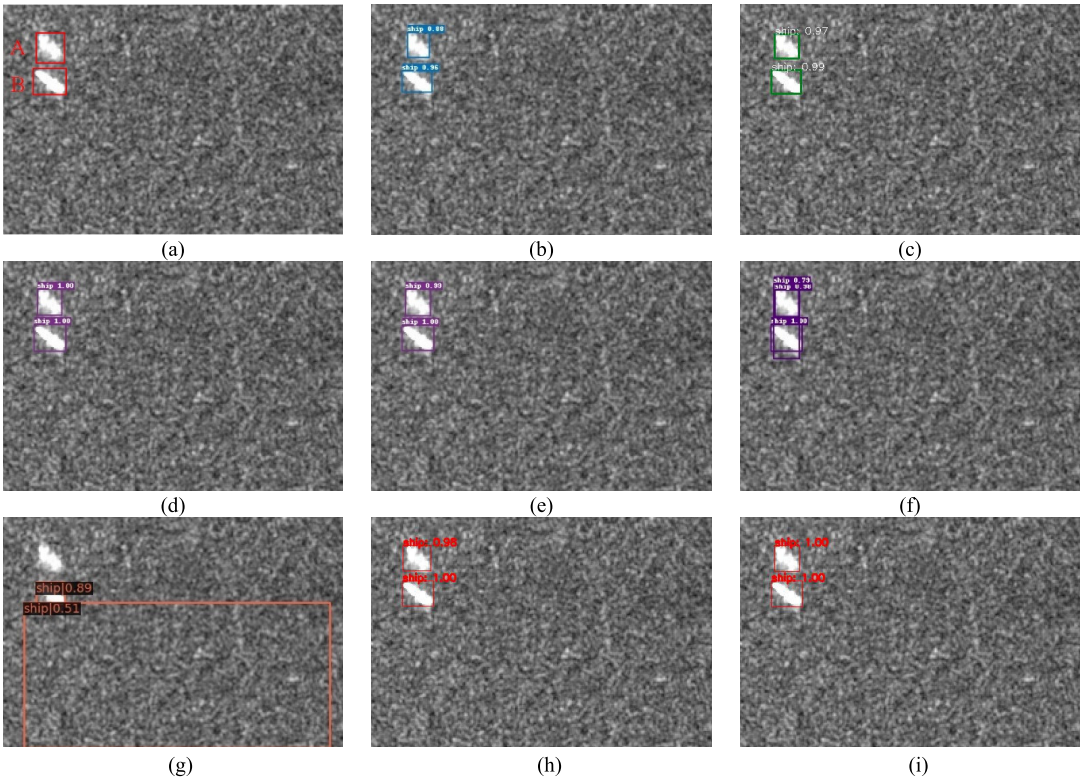


Fig. 12. Object-level detection results of different methods for offshore targets on SSDD: (a) ground truth; (b) YOLOv3; (c) RetinaNet; (d) Faster R-CNN; (e) Cascade R-CNN; (f) HR-SDNet; (g) Yolact; (h) Mask R-CNN; and (i) FINet.

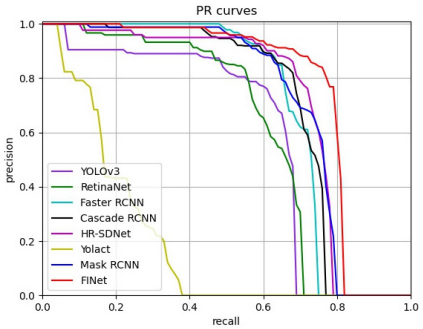


Fig. 13. PR curve of object-level detection methods for inshore targets on SSDD.

SSDD dataset. The one-stage detection algorithm Yolact has the worst results, and the accuracy is only about 50%. The evaluation indexes of the two-stage detection method Mask R-CNN are better, but still lower than FINet. All the indicators of the proposed method are optimal, which shows that FINet can more effectively extract the pixel-level information of the ship target and locate the target more accurately.

The SAR image containing offshore ships is randomly selected from the SSDD dataset to show the detection effect of the pixel-level methods. Fig. 15 illustrates the detection results. Fig. 15(a) shows the original SAR image, Fig. 15(b) shows the ground truth, and Fig. 15(c)–(f) shows the ship detection results of each algorithm. Red circles indicate false detections, and yellow circles indicate missed detections. As can be seen from Fig. 15, VAB has three missed detections and one false detection. SP-CFAR has two missed detections and a large number of false alarms, and Mask R-CNN has one false detection. Only FINet accurately detects nine ships, indicating that its performance in offshore target detection is better than other comparison algorithms.

TABLE VII
EVALUATION INDEXES OF PIXEL-LEVEL DETECTION METHODS FOR INSHORE TARGETS ON SSDD

	Precision	*Recall*	*F1*	AP_{50}	AP_{75}	AP_s
Yolact	0.226	0.415	0.293	0.449	0.232	0.317
Mask R-CNN	0.463	0.513	0.487	0.776	**0.570**	0.512
FINet	**0.483**	**0.544**	**0.512**	**0.848**	0.555	**0.527**

Table VII lists the pixel-level detection evaluation indicators of inshore targets on SSDD. Compared with Table VI, all the indicators of each algorithm have decreased. Yolact still has the worst results, and the accuracy is less than 30%. The accuracy of Mask R-CNN is better than that of Yolact, but still not as good as FINet. The AP_{75} of FINet is only a little lower than Mask R-CNN, but FINet is better than Mask R-CNN in precision, recall, F1 score, AP_{50}, and AP_s. It proves that FINet is suitable for SAR image ship dataset containing a large number of small targets and can more effectively extract the pixel-level information of the ship targets.

We select an SAR image from the SSDD dataset for visualization to show the pixel-level detection effect of inshore targets. The SAR image is shown in Fig. 16. There are

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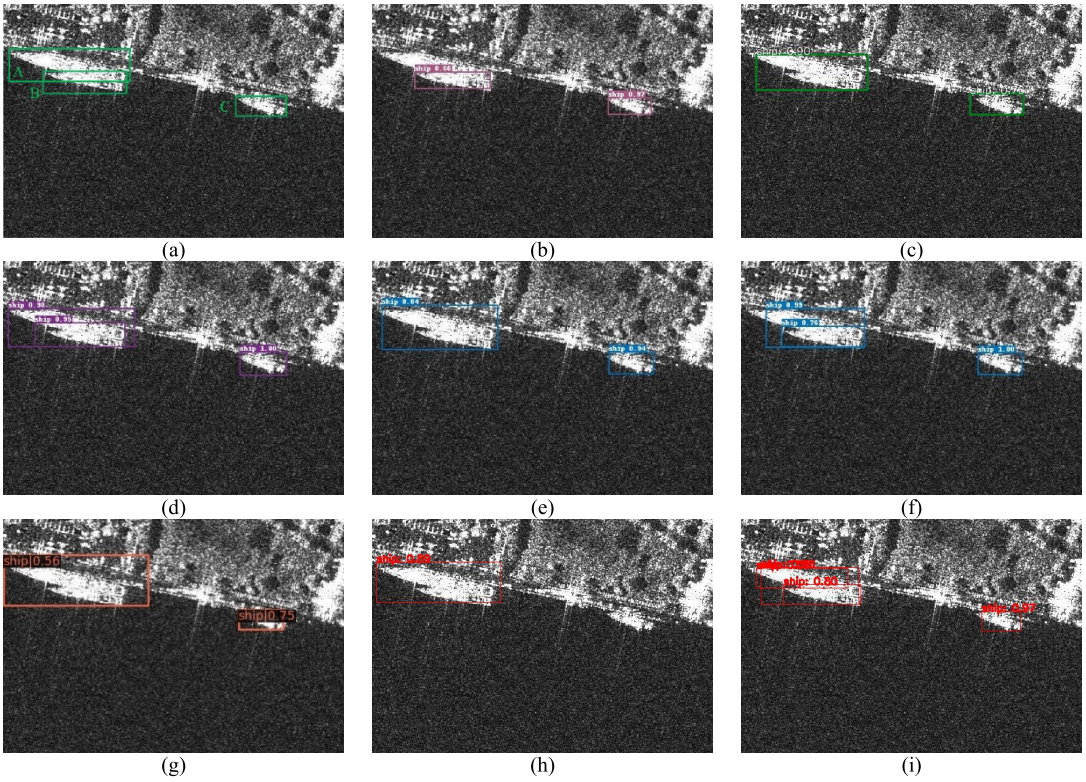


Fig. 14. Object-level detection results of different methods for inshore targets on SSDD: (a) ground truth; (b) YOLOv3; (c) RetinaNet; (d) Faster R-CNN; (e) Cascade R-CNN; (f) HR-SDNet; (g) Yolact; (h) Mask R-CNN; and (i) FINet.

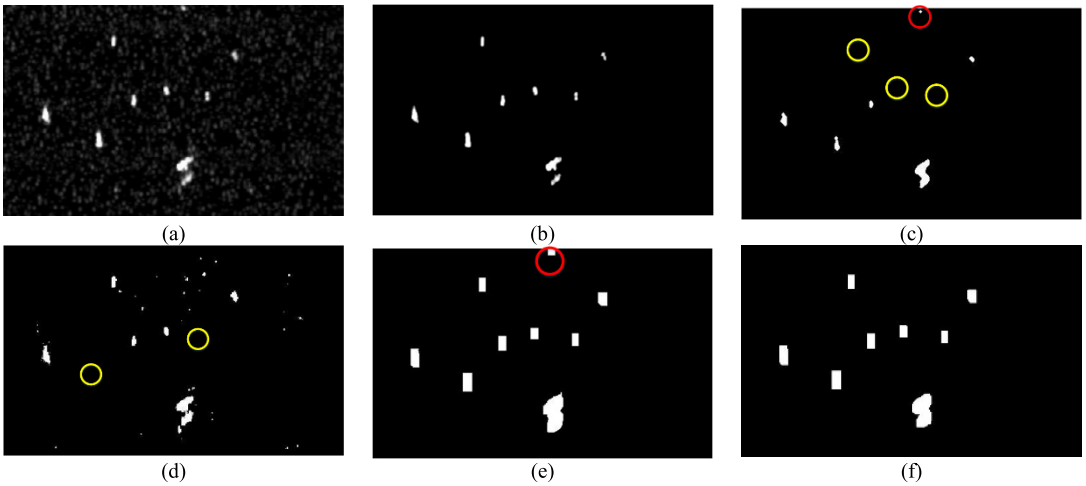


Fig. 15. Pixel-level detection results of different algorithms for offshore targets on SSDD: (a) original image; (b) ground truth; (c) VAB; (d) SP-CFAR; (e) Mask R-CNN; and (f) FINet.

only two inshore ships, but there is a lot of land clutter. Fig. 16(a) shows the original SAR image, Fig. 16(b) shows the ground truth, and Fig. 16(c)–(f) shows the ship detection results of each algorithm. Red circle indicates false detections. Fig. 16 shows that VAB is interfered with land clutter and cannot distinguish the ship target. SP-CFAR cannot completely eliminate land interference, and the ship is partially missing. There is one false detection in Mask R-CNN, while FINet accurately detects two targets. It shows that the ship detection performance of FINet is better than the other comparison algorithms in inshore images, and achieves a satisfactory detection effect.

F. Some Discussion

The target detection algorithm is mainly divided into one-stage detection and two-stage detection. The one-stage detector has great advantages in speed. However, due to its simple feature extraction and recognition strategy, the detection accuracy may be sacrificed (such as YOLOv3 and RetinaNet). The

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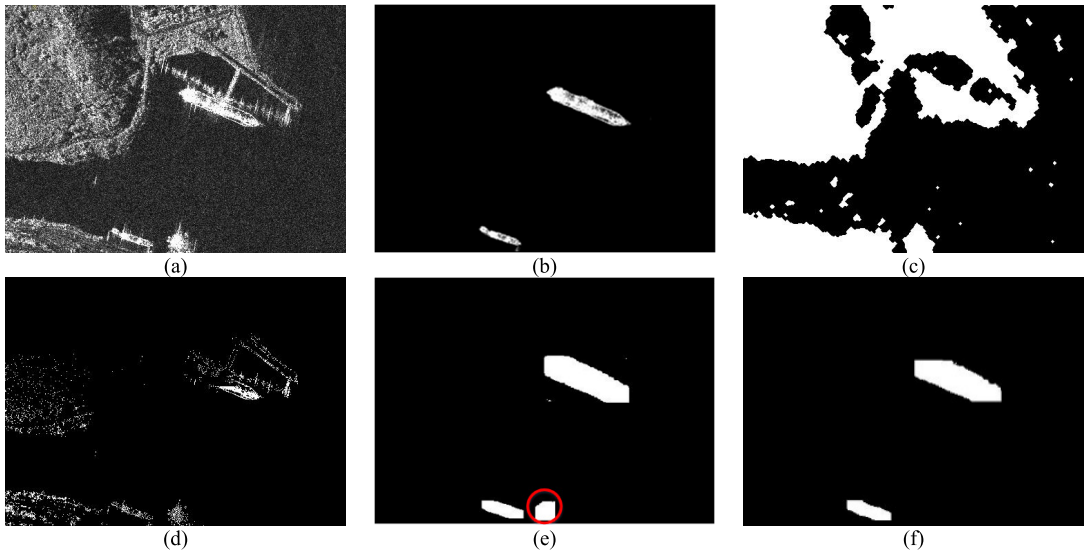


Fig. 16. Pixel-level detection results of different algorithms for inshore targets on SSDD: (a) original image; (b) ground truth; (c) VAB; (d) SP-CFAR; (e) Mask R-CNN; and (f) FINet.

two-stage detector can obtain high detection accuracy due to its well-designed feature representation module. However, due to a large number of ROI proposals and complex processing schemes, their processing speed is low (such as Faster R-CNN and Mask R-CNN). The proposed FINet is improved on the classic two-stage detection algorithm—Mask R-CNN. Because of both object-level and pixel-level detection results and well-designed feature representation module, these two algorithm models are complex, with high detection accuracy and slow speed. In the future, we will deeply study the advantages and disadvantages of one-stage and two-stage detection algorithms to improve the accuracy without sacrificing efficiency.

In Fig. 15, we can observe that the detection results of Mask RCNN and FINet of smaller ships are similar to rectangle, while the detection results of the larger two ships are closer to the shape of the ship. A similar situation occurs in Fig. 16. The two ships in Fig. 16 are larger, so the detection results of Mask RCNN and FINet are close to the shape of the ship. It can be inferred that the detection difficulty of small targets is higher than that of large targets, and the network is more difficult to learn the feature of small targets. Therefore, the detection result of small target is close to rectangle (bounding box) rather than actual shape. In the future, we will further improve the algorithm to enable the network to learn more effectively the feature of smaller targets.

The proposed algorithm and comparison algorithms are tested on HRSID and SSDD datasets. These datasets are composed of images. Therefore, our research mainly focuses on the classification and positioning of ships in the image. Moving ships and ships with waves exist in video, and correlation detection algorithms mainly involve classification, positioning, and tracking. Since the video is composed of continuous images, our algorithm can also be applied to the classification and positioning of ships in the video, providing ideas for the optimization of related algorithms.

Compared with optical data, the special imaging mechanism of SAR makes the targets in SAR data have strong scattering characteristics different from optical data, and the target samples are difficult to obtain. The phase information of a single SAR image has no statistical characteristics, and only the amplitude information can be used for target detection and recognition. And the amplitude information is deeply affected by noise. Optical image imaging is simple and easy to obtain. Optical images usually contain gray information of multiple wavebands, and the noise interference is small. Therefore, SAR images and optical images have both connections and differences. In essence, they all contain gray information and can be used for feature extraction by deep learning. Therefore, we believe that this method can be applied to optical data. However, the specific detection effect needs further experimental verification.

IV. Conclusion

In this study, we present a novel ship detection method based on FINet to improve the recognition accuracy and positioning accuracy of ships in SAR images. FINet contains object-level and pixel-level two branches. The object-level branch is used to obtain object-level detection results, including classification accuracy and positioning bounding boxes. The pixel-level branch is used for pixel-level detection to obtain binary classification results. These two branches conduct multiscale and multitask information interaction and fusion by the designed FIM and FGM. The experimental results on HRSID and SSDD show that, compared with the classical traditional and deep learning detection algorithms, FINet has higher object-level classification precision and more accurate bounding box coordinates. The pixel-level detection can more effectively extracts the pixel-level information of the target and improves the detection accuracy. In the future, we will further research multitask learning for better ship detection performance.

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